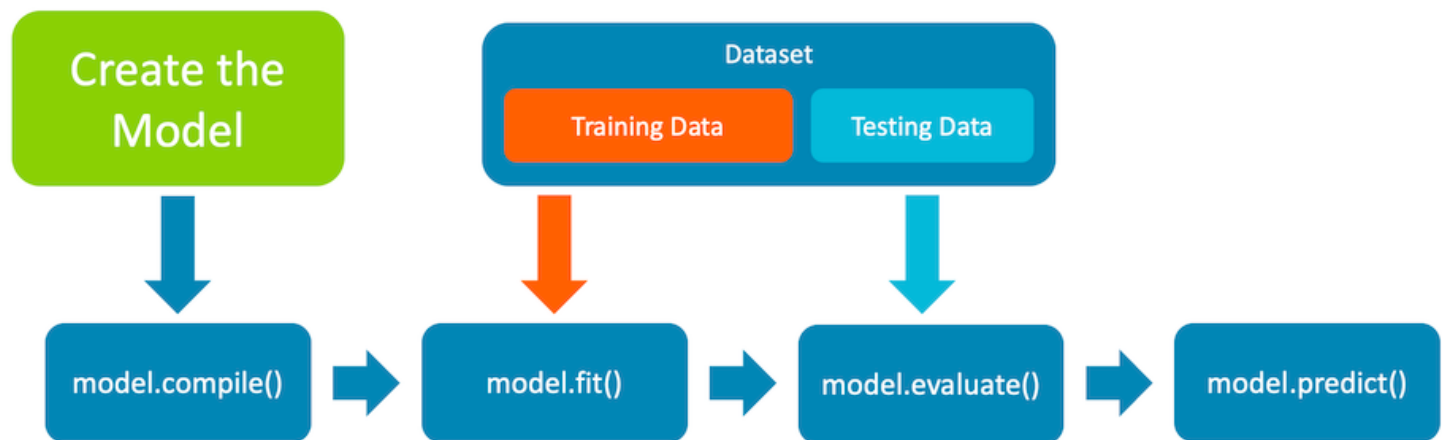


Neural Networks with Keras Part 2

2026.05.14

Keras



How to set up hidden layers?

- Funnel
- Flat
- Expanding
- Non-monotonic

Network shape vs capacity

16 → 8 → 3 (funnel)

12 → 12 → 3 (flat)

8 → 16 → 3 (expanding)

Network capacity

$4 \rightarrow 2 \rightarrow 3$ (small)

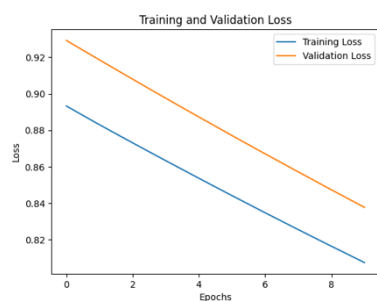
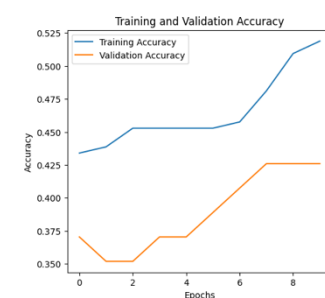
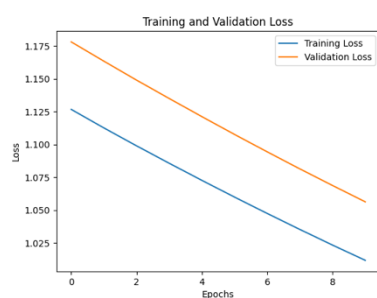
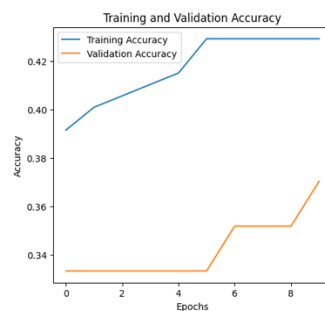
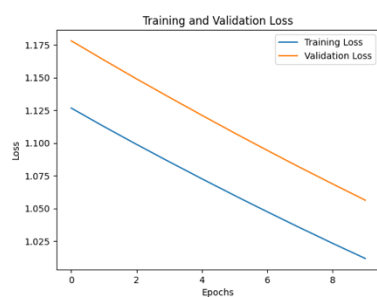
$16 \rightarrow 8 \rightarrow 3$ (medium)

$64 \rightarrow 32 \rightarrow 16 \rightarrow 3$ (large)

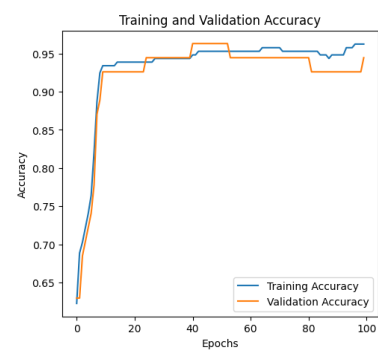
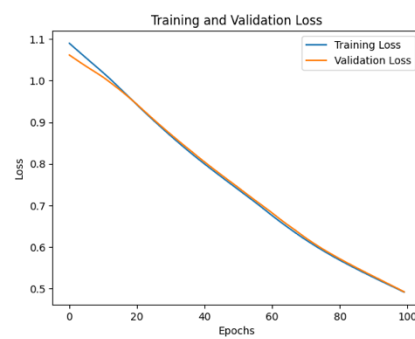
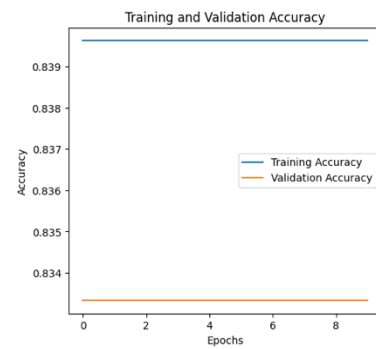
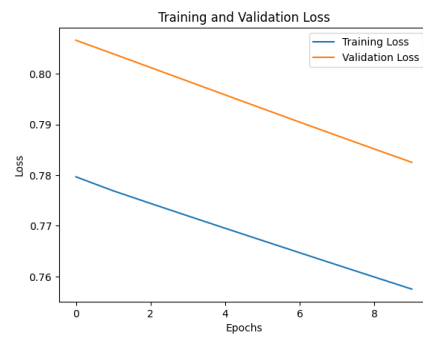
How to determine best capacity?

- 1–2 hidden layers
- 8–64 neurons per layer
- funnel shape, e.g. 32 → 16 or 16 → 8

Unfair model comparisons



Comparing 10x10 vs 100 epochs



Validation split size

Dataset size	Validation split
small	0.2–0.3
medium	0.1–0.2
large	0.05–0.1

Controlling training output

Verbose	model.fit()	model.evaluate()
0	• No output	• Usually no output (may vary)
1 (default)	• Progress bar • Batch updates • Final metrics per epoch	• Progress bar • Summary (often one line for small data)
2	• One clean line per epoch	• One clean summary line • Progress bar

Adjust epochs or capacity?

- Both!

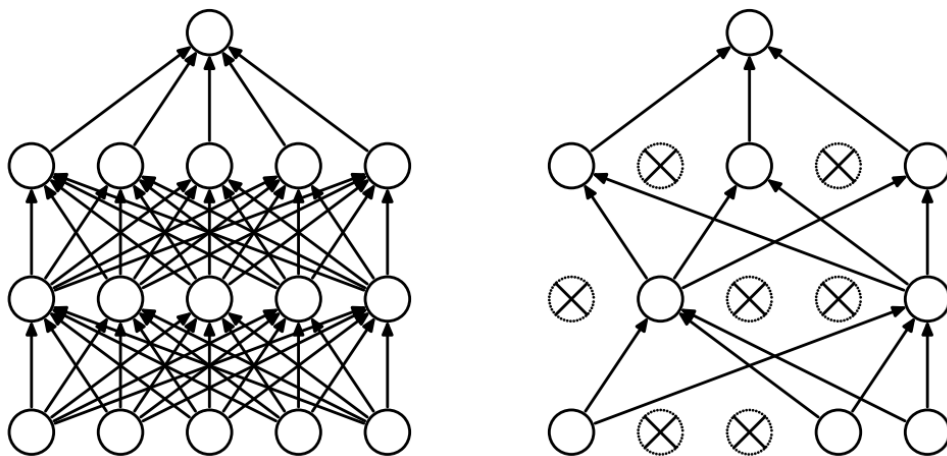
Other regularization methods

- **L2 (weight penalty)**

```
layers.Dense(16, activation="relu",  
             kernel_regularizer=keras.regularizers.l2(0.01))
```

- **Dropout**

```
layers.Dense(16, activation="relu")  
layers.Dropout(0.2)
```



Use cross validation?

Where randomness appears

Data splitting/shuffling

NN initialization/training

Random seed impacts

During development

Final training (production)

NN workflow

- Understand the problem—is NN appropriate?
- Prepare the data
- Simple baseline model
- Train/validation curves
- Adjust capacity (early stopping)
- Fair comparisons
- Tune systematically
- Evaluate final model performance and stability